

A Written de Sitter–Unruh Modified-Inertia Action (v2)

The worldline sector verified, the Cassini quadrupole evaded, MOND shown not to force modified gravity — and the natural covariant (Einstein-aether) completion walled by strong coupling at the GW170817 corner

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Abstract

The de Sitter–Unruh reading of the MOND acceleration scale, $a_0 = cH_\Lambda/Z = c^2\sqrt{(\Lambda/32\pi)} = 9.36 \times 10^{-11} \text{ m s}^{-2}$, posits that the scale is a modified-**inertia** effect of the cosmic (dark-energy) horizon rather than a modification of gravity. Prior work characterized the *obstructions* to a covariant completion — a trichotomy (local $\rightarrow$ Ostrogradsky ghost; field $\rightarrow$ modified gravity failing Cassini; nonlocal $\rightarrow$ the only ghost-free horn) and an all-orders sign wall (a passive, unitary de Sitter bath sources the *anti*-MOND sign). This paper writes the **constructive** counterpart — an explicit nonlocal modified-inertia action — and carries its covariant analysis to a definite boundary. Four results are established, each with reproducible verification (both a_0 footings): **(1) the worldline / matter sector is written and machine-verified** ($v(y)=\sqrt{(1+1/y)}$ to 3×10^{-13} , Newton, deep-MOND, baryonic Tully–Fisher, the external-field kernel with a dS-Unruh-forced $\sqrt{2}$ DC weight, exact ghost-freedom, conservativeness); **(2) the modified-inertia realization evades the Cassini quadrupole bound** by ~ 7 orders of magnitude (deep-Newtonian $v-1 \approx 7 \times 10^{-7}$ suppression), where the modified-gravity realization fails it at $+6$ to $+14\sigma$; **(3) promoting the frame field to a covariant Einstein-aether field, MOND does not force modified gravity** — the matter source is $l=0$ (parallel to the frame field, soaked by the unit-timelike multiplier), evading the Bianchi $l=2$ shear lock that kills the modified-gravity limb, and the Cassini-inert corner is a genuinely *open* region; **(4) the mixed matter–aether two-point propagator is ghost-free in closed form** (an all-orders resummation, single healthy pole), with no resonant un-suppression of the Cassini residual. The completion is nonetheless **not** a finished field theory along this route: the observationally-required corner — GW170817 ($c_{13} \lesssim 10^{-15}$) intersected with the preferred-frame safety the inertia-evasion needs ($\alpha_1=\alpha_2=0$) — is the $c_{13}=c_{14}=0$ double-null where the Einstein-aether kinetic norms vanish, a **strong-coupling / ill-posed point**; the linear analysis loses control there and the nonlinear strong-coupling window ($M_{\text{sc}} \sim 10^{11} \text{ GeV}$, *not excluded* but undemonstrated) is the open question. The MOND sign remains a **postulate** (walled all-orders) and a_0 's value is underived. This is an effective completion carried to a sharp boundary — **not a finished theory and not a theory of everything**.

1. Setting and scope

The framework reads a_0 as the acceleration at which a body's inertial response to the de Sitter vacuum departs from the Newtonian value, with the Deser–Levin temperature $T(a) =$

$(\hbar/2\pi k_B c)\sqrt{(a^2 + (cH_\Lambda)^2)}$ a horizon floor and the interpolation $g_{\text{obs}} = \sqrt{(g_{\text{bar}}^2 + g_{\text{bar}} a_0)}$, $v(y) = \sqrt{(1+1/y)}$, $y = g_{\text{bar}}/a_0$.

Two facts set the task. A covariant **modified-gravity** realization of this scale (AeST [Skordis–Złótnik 2021], the Blanchet–Skordis khronon [2025]) inherits a solar-system quadrupole tension with the improved Cassini bound $Q_2 = (1.6 \pm 1.8) \times 10^{-27} \text{ s}^{-2}$ [Park et al. 2026] at $+6$ to $+14\sigma$ for the framework’s own slow interpolation. A modified-**inertia** realization instead evades that bound (§3), because in modified inertia the Sun’s field stays Newtonian — no phantom-density quadrupole. The inertia realization is therefore the one to write down; this paper writes its worldline sector and then tests whether it lifts to a well-posed covariant field theory. The framework’s earlier theory-of-everything and Standard-Model claims were publicly retracted (2026-06-23) and are not reasserted; what follows is a one-scale effective-theory construction with named inputs, each labeled **derived**, **postulated**, or **open**.

2. The action

With signature $(-+++)$ and c restored, $S[x, g, u] = S_{\text{EH}}[g] + S_{\text{u}}[g, u] + S_{\text{matter}}[x, g, u]$:

- **Host gravity, unmodified:** $S_{\text{EH}} = (1/16\pi G) \int d^4x \sqrt{-g} (R - 2\Lambda)$.
- **Frame constraint** — a unit-timelike clock u^μ with a Lagrange multiplier, and (*in the worldline sector*) no Maxwell-type kinetic term: $S_{\text{u}} = - \int d^4x \sqrt{-g} (\lambda/2) (u^\mu u_\mu + 1)$.
- **Modified-inertia content**, in the matter kinetic sector, ρ_{m} -weighted and gated by a nonlocal branch-cut form factor:

$$S_{\text{matter}} = - \frac{1}{2} \int d^4x \sqrt{-g} \rho_m [s u^\mu K(\square_u/a_0^2) u_\mu], \quad K(z) = \frac{\sqrt{1+4z} - 1}{2\sqrt{z}}, \quad \square_u f = u^a \nabla_a (u^b \nabla_b f).$$

K is non-polynomial with a branch cut and a single healthy pole (residue $+1$); this nonlocality is the ghost-free horn of the trichotomy. The equivalent Galley doubled-worldline form uses the conservative gradient $G(a)^\mu = a^\mu \mu_{\text{fw}}(|a|/a_0)$, μ_{fw} the exact inverse of $v(y) = \sqrt{(1+1/y)}$. The external-field kernel $\theta(y) = \theta_0/(1+(\theta_0-1)y^2)$, $\theta_0 = \sqrt{2}$, has its Lorentzian form-class and $\sqrt{2}$ DC weight forced by the de Sitter worldline Wightman function $W(u) \propto -1/\sinh^2(\kappa(u-i\epsilon)/2)$. Here **$s = -1$ is the MOND-sign postulate**, $a_0 = cH_\Lambda/Z = 9.36 \times 10^{-11} \text{ m s}^{-2}$ (canonical, ρ_{DE} , $Z = \sqrt{(32\pi/3)}$), alternate footing $1.13 \times 10^{-10} (\rho_{\text{total}})$.

3. Verified: the worldline sector, and the Cassini evasion

Machine-checked, both footings (§8): (i) the circular-orbit reduction $\mu_{\text{fw}}(a/a_0)a = g_{\text{N}}$ reproduces $v(y) = \sqrt{(1+1/y)}$ to 3.35×10^{-13} ; (ii) Newton and deep-MOND limits; (iii) baryonic Tully–Fisher $v^4 = GM a_0$ (exact); (iv) the identity inverting $Y = \mu_{\text{fw}}(X)X \rightarrow X = \sqrt{(Y^2 + Y)}$, i.e. $g_{\text{obs}} = \sqrt{(g_{\text{bar}}^2 + g_{\text{bar}} a_0)}$; (v) the external-field kernel with the $\sqrt{2}$ DC weight; (vi) **ghost-freedom** — $K(z)$ has a single pole (residue $+1$), the local truncation an Ostrogradsky ghost (residue -1); (vii) conservativeness.

The Cassini evasion. In modified inertia the Sun’s field stays Newtonian; the anisotropic galactic external field ($a_{\text{ext}} = 2.29 a_0$) enters only Saturn’s inertial response, deep-Newtonian-suppressed by $v-1 = a_0/2a \approx 7 \times 10^{-7}$ at Saturn ($y = 7 \times 10^5$). A Legendre decomposition gives an $l=1$ dipole $\sim 1.5 \times 10^{-28} \text{ s}^{-2}$ (first order in a_{ext}) and the $l=2$ quadrupole $\sim 7.4 \times 10^{-34} \text{ s}^{-2}$ (second order) — the latter, which Cassini constrains, ~ 7 orders below the 5.2×10^{-27} ceiling. The modified-gravity

realization, whose MOND is carried *by* the aether twist, instead fails at $+6$ to $+14\sigma$. This is the derived realization of Milgrom's [2009] statement that the inner-solar-system Q_2 anomaly is an AQUAL/QUMOND (modified-gravity) effect absent in modified inertia.

4. The covariant completion: two wins and a wall

We promote u^u to a covariant Einstein-aether field with the general two-derivative kinetic term $L_{\text{kin}} = -M^2[c_1(\nabla u)^2 + c_2(\nabla \cdot u)^2 + c_3 \nabla_\mu u_\nu \nabla^\mu u^\nu + c_4 a^2]$ [Jacobson–Mattingly 2001], and test whether the completion is a well-posed field theory.

Win 1 — MOND does not force modified gravity. Varying S_{matter} , the source it adds to u^u 's field equation is $J^u = -\rho_m s^k u^u$: **$l=0$, parallel to u^u** , soaked entirely by the unit-timelike multiplier λ . It does not populate the $l=2$ traceless-shear channel whose nonzero divergence ($\nabla \cdot \text{shear} = (2/3)\nabla^\mu \nabla^\mu f$, sympy-exact) is the Bianchi lock that forces AeST's Φ -sourcing pressure and its Cassini quadrupole. The framework's MOND, living in S_{matter} , evades that lock. A plain inert aether carries at most $Q_2^u \lesssim 7 \times 10^{-35} \text{ s}^{-2}$ (both footings), ~ 8 orders under the bound, over a genuinely **open, ghost-free, no-Cherenkov region** of (c_1, c_3) — not a measure-zero tuning. So the covariant lift does **not** collapse to modified gravity.

Win 2 — the mixed propagator is ghost-free in closed form. Varying the *fully nonlocal* K with u^u dynamical (u^u inside $\square_u$) generates an infinite higher-derivative source tower that **resums**: $Q(z) = K + 2zK' = 2\sqrt{z}/\sqrt{1+4z}$. The mixed matter–aether two-point propagator has a single $+1$ (healthy) pole, $\text{Re } \Sigma = 0$ on-band, and the branch cut is a radiation continuum, not a pole. This is all-orders — verified against the closed-form resummation, not a truncation — and the dynamical- u coupling does **not** resonantly un-suppress the $(v-1)^2$ -suppressed Cassini residual.

The wall — strong coupling at the observationally-required corner. Every ghost-free / hyperbolic / no-Cherenkov certification above was obtained at $c_{13} = c_1 + c_3 = 0.787$ — a point where the spin-2 graviton travels at $2.17c$, **excluded by GW170817 by ~ 15 orders** ($|c_{13}| \lesssim 10^{-15}$ [Gong et al. 2018]). On the framework's own corner, $c_{14} \propto c_{13}$, and the preferred-frame safety the inertia-evasion needs ($\alpha_1 = \alpha_2 = 0$) independently forces $c_{14} = 0$ at $c_{13} = 0$. The allowed corner is therefore the **$c_{13} = c_{14} = 0$ double-null**, where the Einstein-aether kinetic norms vanish and the spin-1 speed diverges — the theory's known strong-coupling / ill-posed point. The nonlocal MI matter coupling is a_0 -IR-gapped ($dQ/dz \rightarrow 0$ for $\omega \gg a_0$); it renormalizes the kinetic norm only in the galactic infrared and supplies nothing in the ultraviolet where the strong coupling lives. The linear two-point analysis loses control at this corner; the strong-coupling scale is high ($M_{\text{sc}} \sim 10^{11} \text{ GeV}$, so a controlled nonlinear/Stückelberg window is *not* excluded), but that window must be **demonstrated, not assumed**. Along the natural Einstein-aether route the completion is **walled at the linear level**.

5. What is postulated, and what remains open

Postulated (flagged, never used as results): the **MOND sign** $s=-1$ — the passive, unitary de Sitter bath gives $\delta m = 2\int \rho(\omega)/\omega^2 d\omega \geq 0$ (anti-MOND) at every computable order (2-point spectral, all-orders Bisognano–Wichmann, ground-state passivity, the Euclidean-KMS corner, and the connected 4-point positivity), so the MOND sign requires an active/pumped medium de Sitter

does not supply; it is admitted only through a named Machian posit. The **value** $Z=\sqrt{(32\pi/3)}$ (only the scale $a_0\sim\sqrt{\Lambda}$ is forced) and the corner ω_c .

Open: (i) the nonlinear/Stückelberg strong-coupling analysis at the $c_{13}=c_{14}=0$ double-null — whether a controlled EFT window below M_{sc} survives (the $c_{13}=0$ plane is a live degeneracy in the aether literature; a window is not excluded); (ii) whether a *different* frame-field structure (beyond a two-derivative vector) evades the double-null without becoming the modified-gravity mimetic limb; (iii) the off-circular / congruence-shear terms in $\square_u$ beyond the constant- $|a|$ slice; (iv) the separate four-point / de Sitter-positivity vertex.

6. Honest standing

The inertia-based worldline sector is written and verified; the modified-inertia realization evades Cassini; MOND does not force modified gravity; and the mixed two-point propagator is ghost-free in closed form. These are real, both-footing results. The completion is nonetheless **not** a finished field theory along the natural Einstein-aether route: the corner it is required to occupy — GW170817 $\cap$ preferred-frame safety — is the aether strong-coupling double-null, where the linear analysis breaks down and the nonlinear window is undemonstrated (though not excluded). The MOND sign stays postulated and a_0 's value underived. This paper therefore reports a sharp boundary: the covariant modified-inertia completion is neither the modified-gravity limb (Cassini-failing) nor a finished field theory — it is a well-posed worldline theory whose covariantization is walled at a specific, named, literature-live strong-coupling point. It makes no completeness or theory-of-everything claim.

7. Related work

The construction sits between the relativistic modified-gravity realizations of MOND — AeST [Skordis & Złośnik 2021], the Blanchet-Skordis khronon [2025], and the nonlocal (mimetic modified-gravity) model of Deffayet & Woodard [2026] — and the modified-inertia program of Milgrom [1999, 2009]. The covariant frame-field sector is Einstein-aether [Jacobson & Mattingly 2001; Jacobson 2008], whose PPN structure [Foster & Jacobson 2006] and GW170817 constraint [Gong et al. 2018] supply the wall. The nonlocal form factor's ghost-freedom is of the class analyzed by Barvinsky [2003]; the doubled-worldline nonconservative formalism is Galley [2013].

8. Reproducibility

All numerical and symbolic claims are backed by committed scripts (repository `real_research/`): the worldline sector and Cassini evasion in `reviews/cassini_mi_evasion_2026/` and the integrated check; the covariant kinetic wins in `reviews/mi_kinetic_completion_2026/` ($l=0$ source, open inert corner, both footings); the mixed-propagator ghost-freedom and the strong-coupling wall in `reviews/mi_propagator_2026/` (`propagator_compute.py`, `refute_strongcoupling.py`). The sign-wall inputs are `ACTIVE_KERNEL_SIGNTHEOREM_2026-06.md` and `FOURTH_ORDER_SIGN_WALL_2026-07.md`; the trichotomy is `COVARIANT_MI_COMPLETION_2026-06.md`.

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14. Companion (this program): *No Pump-Free Corner* (sign wall, all-orders), Zenodo 10.5281/zenodo.21184373; the covariant-MI quadrichotomy, Zenodo 10.5281/zenodo.21148494; the written MI action v1, Zenodo 10.5281/zenodo.21253645.

Both a_0 footings carried throughout. Each claim is labeled derived / postulated / open; the MOND sign is a postulate; the covariant completion is walled at a named strong-coupling corner with the nonlinear window open. No completeness or theory-of-everything claim is made.